

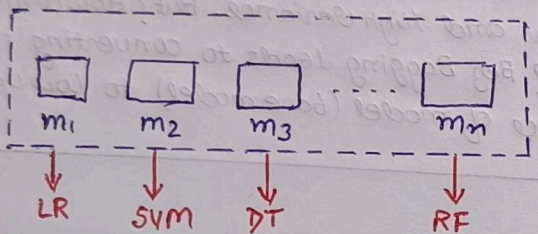
Ensemble Learning

Intro. to Ensemble Learning:

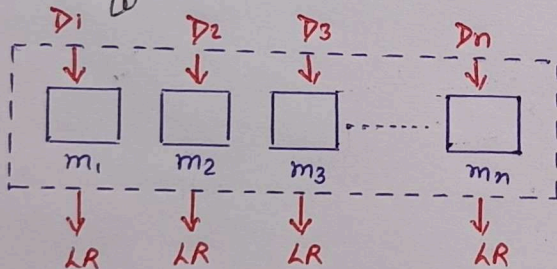
⇒ What is the wisdom of the crowd? It's the concept that the collective opinion, judgement or intelligence of a large and diverse group of people often outperforms the knowledge of a single expert.

⇒ The core idea in Ensemble Learning is that it uses different models, this variety can be achieved by:

(i) using actually different models or techniques on the same data.



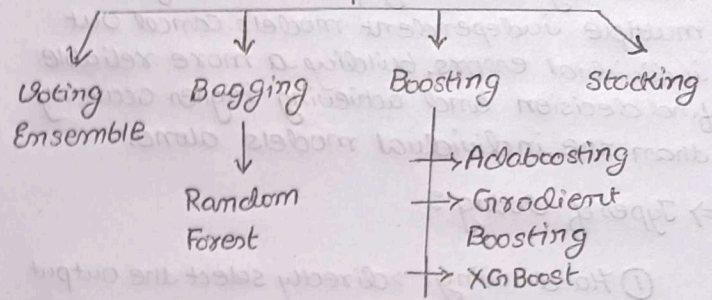
(ii) using same technique but training each with different datasets.



(iii) combination of (i) and (ii).

⇒ How prediction is done in E.L. then? Each model will have its own predictions, we go for the predicted result which be most frequent or be in majority count. For the classification this be implied, while for the regression mean is calculated of each model's output.

⇒ Types of Ensemble Learning:



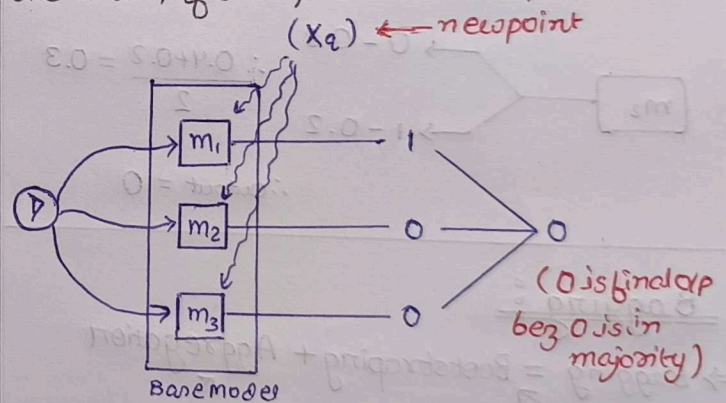
⇒ Why Ensemble Learning works? It reduces the noise of the different models.

⇒ Benefits of E.L.:

- ① Improvement in Performance.
- ② Bias-variance is reduced, leading to low bias and low variance.
- ③ Robustness

Voting Ensemble:

⇒ Here ind. all models are trained on the same data point. The new point is sent to each model, each give own classification prediction. The final be the majority of the prediction, while for regression, final output be the mean of each predicted output.



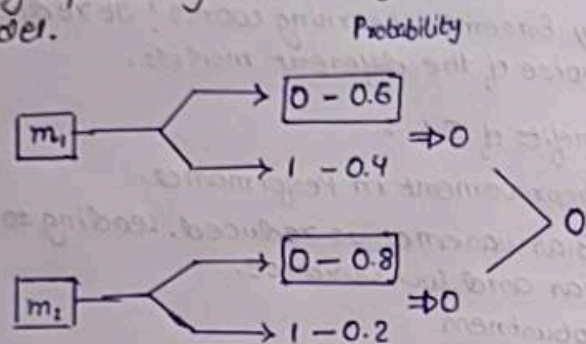
⇒ Assumptions made in voting ensemble:

- ① Models be very independent (or different) from each other.
- ② Each model have atleast 50% accuracy.

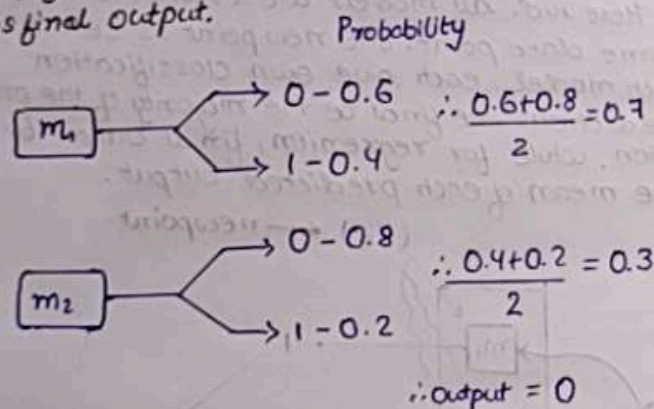
⇒ Why Voting works? A voting ensemble works by combining predictions from multiple independent models cancel out individual errors, building a more reliable final decision and achieving higher accuracy than the individual models alone.

⇒ Types of Voting:

① Hard voting → directly select the output of higher probability as result of that individual model.



② Soft voting → we find the avg of each classified prob. and then find avg the higher one as final output.



Bagging:

⇒ Bagging = Bootstrapping + Aggregation

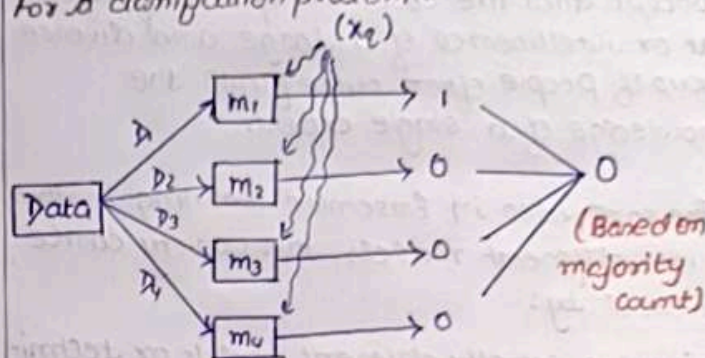
creating multiple random subsets of your original data by sampling with replacement

combining the results of all the models

⇒ Bagging: It's a ML technique that combines multiple models to make more accurate and stable predictions.

In standard bagging, we train the same base algorithm on every single bootstrap subset of the data.

For a classification problem:



⇒ We use model for base model which be Low Bias and High Variance. But? How can it be helpful? By Bagging leads to converting High Variance of model (base model) to Low Variance.

